MA321 Quiz 3 Review Answers

1. % Error = Abs(Difference) ÷ Actual.

Month	Actual	Forecast	Difference	Abs(Difference)	% Error
1	50	45	5	5	10%
2	80	88	−8	8	10%
3	40	50	−10	10	25%

MAPE = (10% + 10% + 25%) ÷ 3 = **15%**

2. Answer True or False for each.

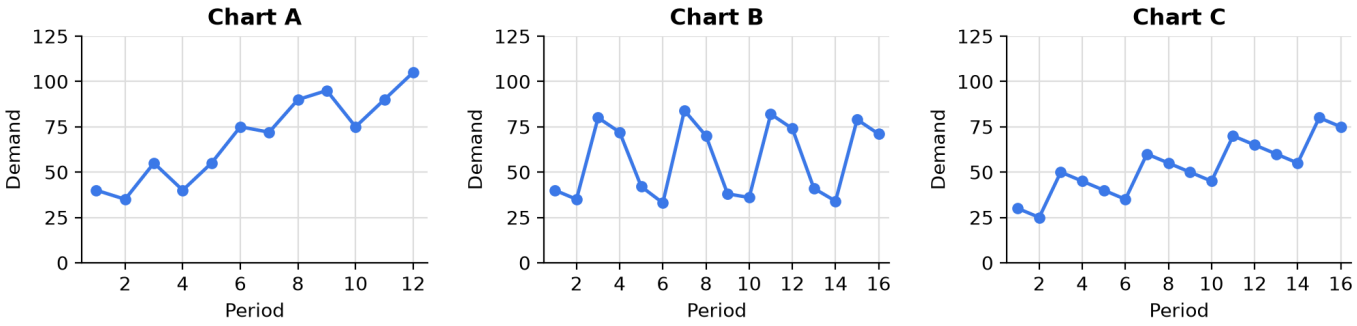


Chart A: upward trend, not seasonal, lots of noise.
Chart B: no trend (stationary), seasonal (repeats every 4 periods), little noise.
Chart C: upward trend, seasonal (repeats every 4 periods), little noise.

a) **True.** Chart B has no trend; it moves up and down around the same level.

b) **False.** Chart A has no repeating pattern; its ups and downs are noise.

c) **True.** Chart C goes up over time.

d) **True.** Chart A jumps around a lot; Chart C follows a clean repeating pattern.

e) **False.** Chart C has an upward trend, so it is not stationary.

3. Forecast = baseline × seasonal index.

a) Baseline 20: 20 × 0.6 = **12**, 20 × 1.2 = **24**, 20 × 0.95 = **19**, 20 × 1.1 = **22**, 20 × 1.4 = **28**, 20 × 0.75 = **15**

b) 20% lower baseline = 20 × 0.8 = 16: **9.6, 19.2, 15.2, 17.6, 22.4, 12** (each forecast in a is 20% lower too)

4.

a) **Forecast 2.** The lower MAPE (12%) is more accurate.

b) The forecast graph stays very close to the actuals graph. On average the forecast is only 3% off from the actual demand.

c) **True.** If the forecasts are far off (for example, forecasting 30 when the actual is 10 is a 200% error), the MAPE can be over 100%.

5. Deseasonalized demand = actual ÷ index.

January: 36 ÷ 0.8 = **45**
February: 54 ÷ 0.9 = **60**

March: $50 \div 1.0 = \mathbf{50}$
April: $60 \div 1.2 = \mathbf{50}$
May: $63 \div 1.5 = \mathbf{42}$
June: $30 \div 0.6 = \mathbf{50}$

b) **Best: February** (60), the highest deseasonalized demand. **Worst: May** (42), the lowest.

May had the highest actual sales (63), but May is expected to be a big month (index 1.5), so it actually underperformed. June had the lowest actual sales (30), but June is a slow month (index 0.6), so it performed at an average level (50).